

Course: Cloud Data Engineering Laboratory Course Code: CSE421

Practical 1

Aim: Create a cloud storage bucket and upload a dataset (CSV/JSON). Demonstrate basic operations such as upload, download, and file organization using Amazon S3/Azure Blob Storage.

Objectives:

- To understand cloud object storage concepts.
- Create storage bucket/container.
- Upload/download and organize files.

Outcomes: Students will create buckets, upload/download datasets and organize files.

Theory: Cloud object storage stores data as objects in buckets/containers for scalable storage.

Input: AWS/Azure account and CSV/JSON dataset.

Output:

Steps:

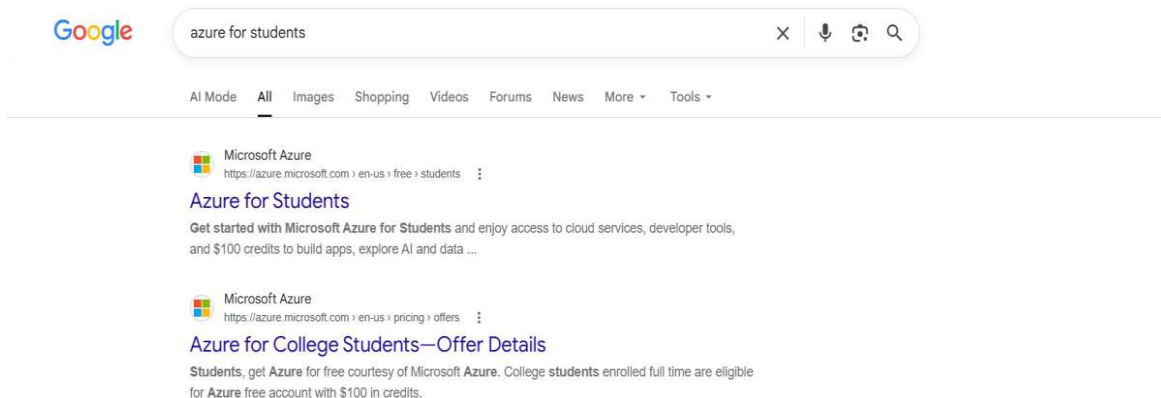
Account Creation, Resource Group, Storage Account, Container Setup & File Upload

This documentation serves as a comprehensive, step-by-step practical lab guide to configure Microsoft Azure cloud services. Follow these instructions to set up an Azure Student Account, create a Resource Group, deploy a Storage Account, create a Blob Container, and upload and verify files in the cloud.

1. Setting up an Azure for Students Account

Azure for Students provides \$100 in free cloud credits and access to popular cloud services for 12 months without requiring a credit card.

- **Step 1.1:** Search for 'azure for students' on Google and select the official Microsoft Azure offering.



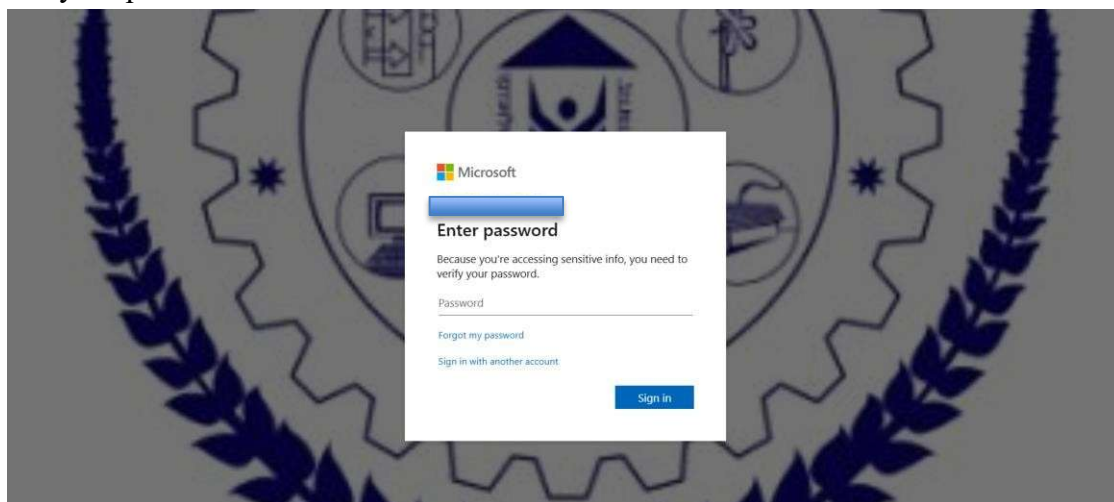
Step Image: Screenshot 2026-07-25 163106.png

- **Step 1.2:** Click on the blue 'Start free' button on the landing page.



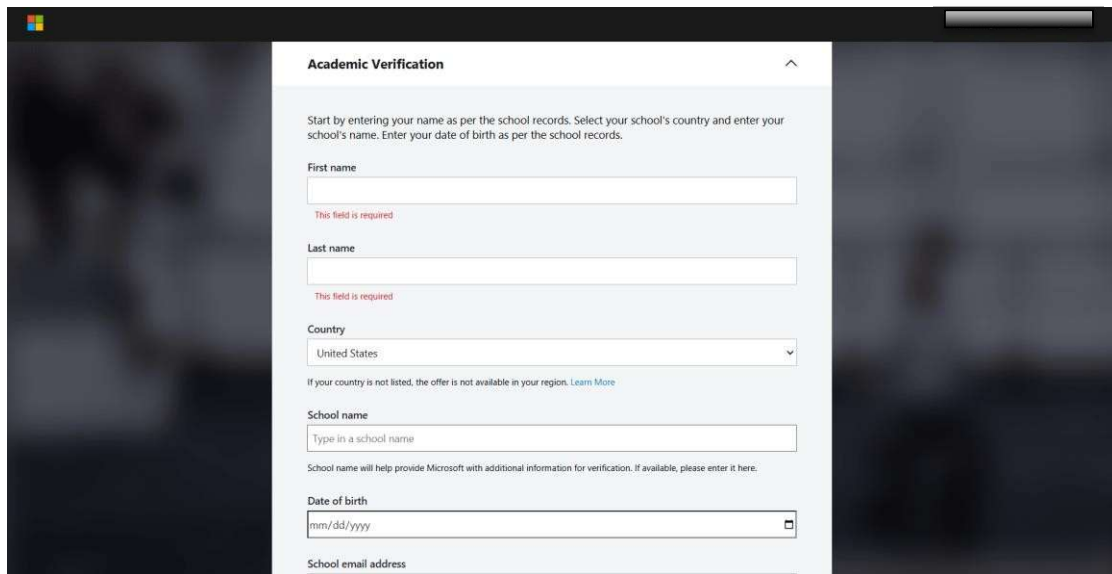
Step Image: Screenshot 2026-07-25 163136.png

- **Step 1.3:** Sign in with your academic organization email (e.g., yourname@mit.asia) and enter your password.



Step Image: Screenshot 2026-07-25 163224.png

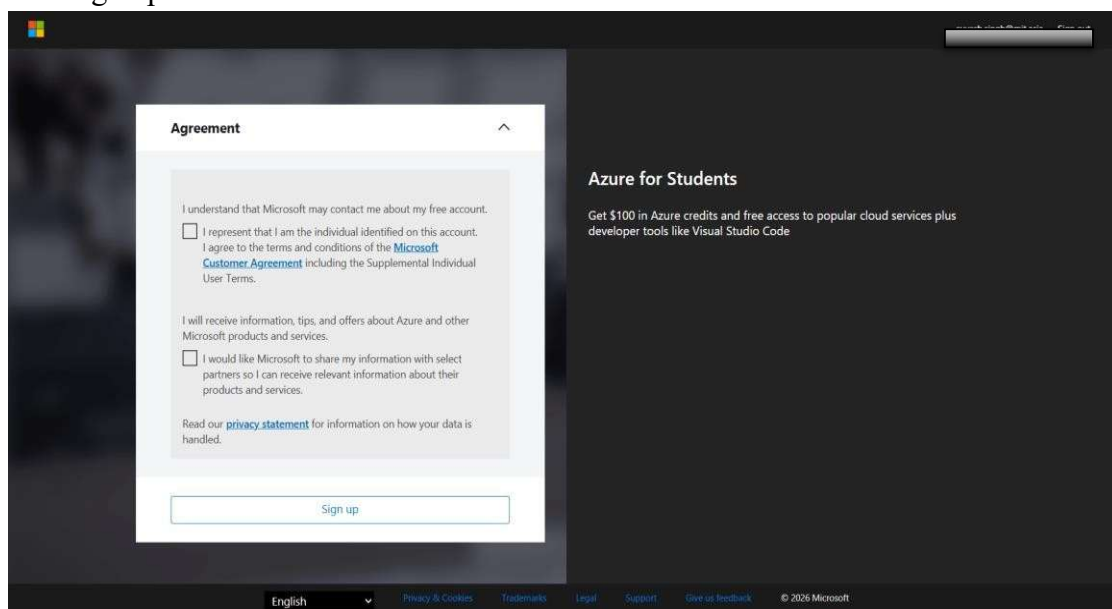
- **Step 1.4:** Complete the Academic Verification form. Fill in details like your name, school, date of birth, and academic email.



The screenshot shows the 'Academic Verification' form. It includes fields for 'First name', 'Last name', 'Country' (a dropdown menu currently showing 'United States'), 'School name', 'Date of birth' (with a date picker), and 'School email address'. Red text below the first and last name fields indicates 'This field is required'. A note below the country dropdown states: 'If your country is not listed, the offer is not available in your region. [Learn More](#)'. Below the school name field, a note says: 'School name will help provide Microsoft with additional information for verification. If available, please enter it here.'

Step Image: Screenshot 2026-07-25 163347.png

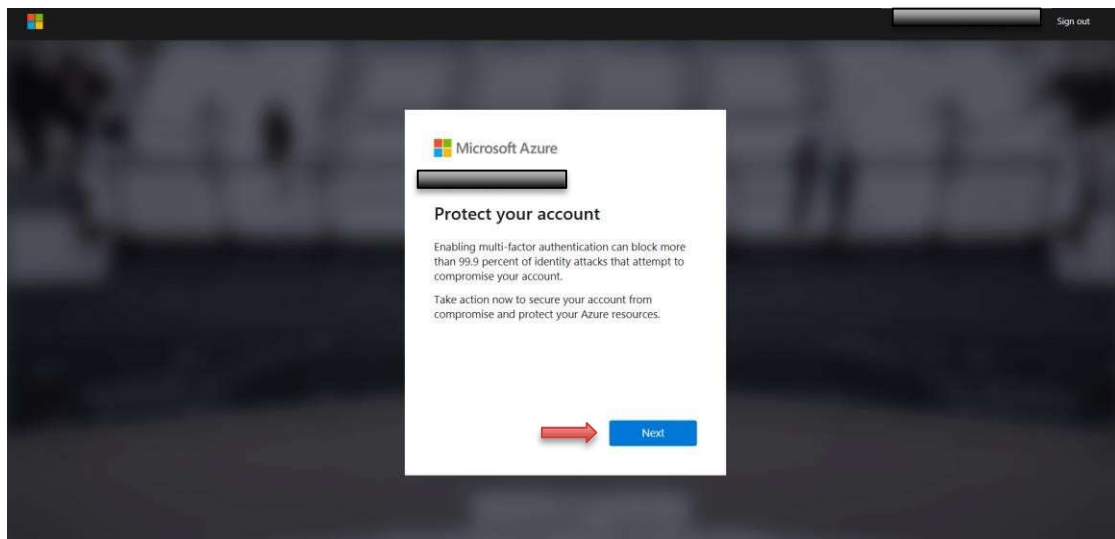
- **Step 1.5:** Review the Microsoft Customer Agreement, check the verification boxes, and click 'Sign up'.



The screenshot shows the 'Agreement' page. It contains two checkboxes: the first is for agreeing to the Microsoft Customer Agreement, and the second is for allowing Microsoft to share information with select partners. Below these is a link to the 'privacy statement'. A 'Sign up' button is at the bottom of the form. To the right, the 'Azure for Students' section offers '\$100 in Azure credits and free access to popular cloud services plus developer tools like Visual Studio Code'. The footer includes a language dropdown set to 'English', links for 'Privacy & Cookies', 'Trademarks', 'Legal', 'Support', 'Give us feedback', and a copyright notice for 2026 Microsoft.

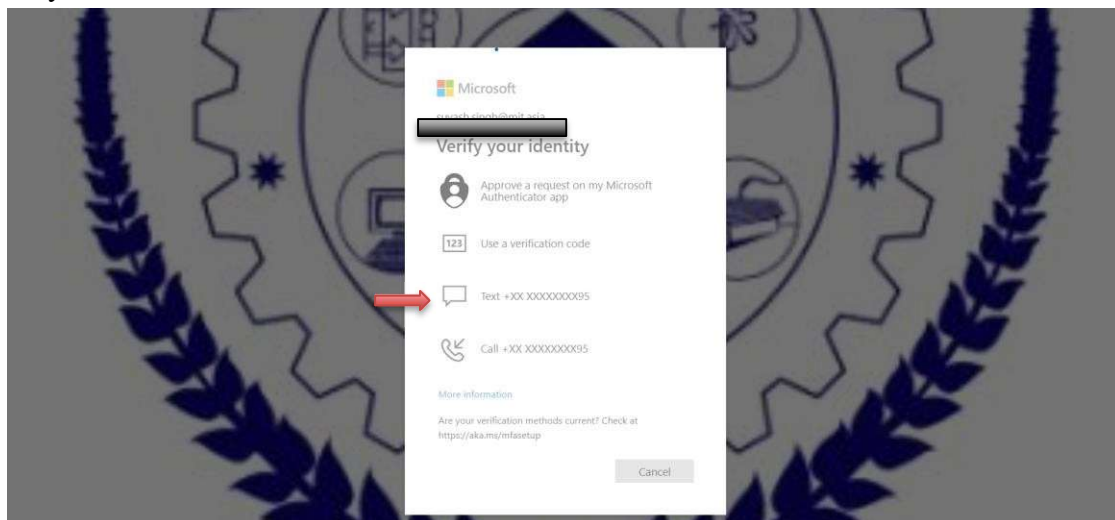
Step Image: Screenshot 2026-07-25 163557.png

- **Step 1.6:** Set up Multi-Factor Authentication (MFA) to secure your account. Click 'Next' on the prompt.



Step Image: Screenshot 2026-07-25 163614.png

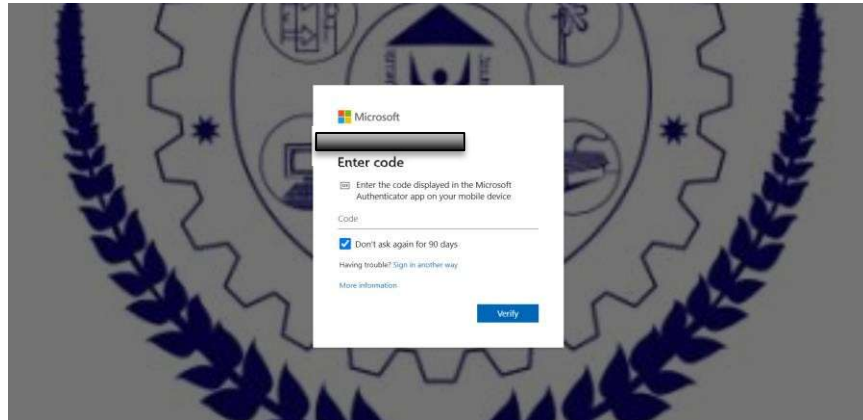
- **Step 1.7:** Choose a verification method: You can use the Microsoft Authenticator App or verify via SMS/Phone Code.



Step Image: Screenshot 2026-07-25 163817.png

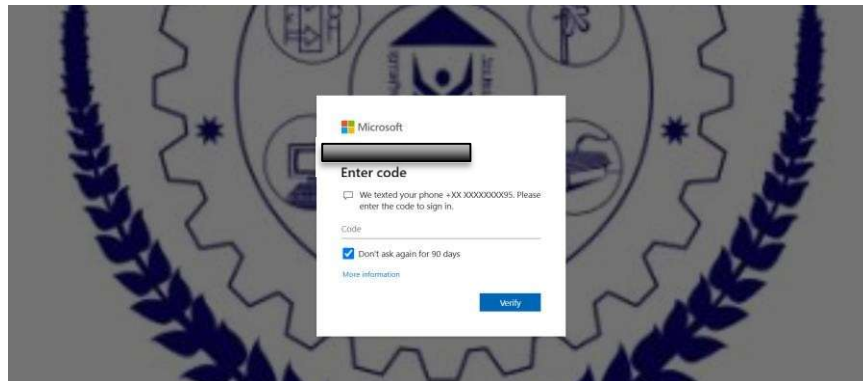
- **Step 1.8:** If using Authenticator, enter the code from your app. If using SMS, enter the verification code texted to your registered phone number.

- Authenticator Verification:



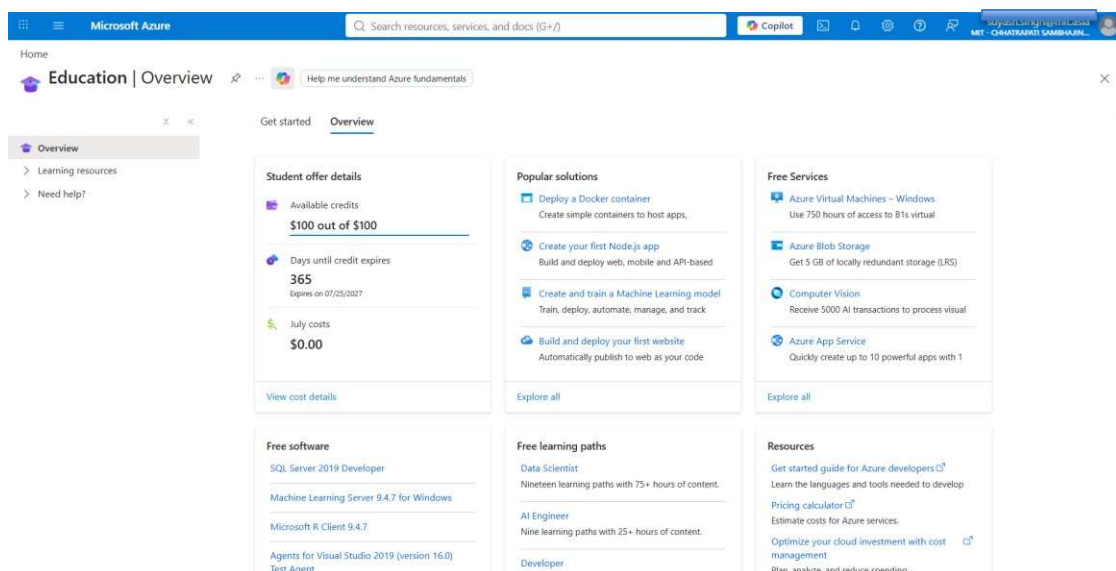
Step Image: Screenshot 2026-07-25 163801.png

- Phone SMS Verification:



Step Image: Screenshot 2026-07-25 163829.png

- **Step 1.9:** Once identity verification is successful, the browser redirects you to the Azure Portal Education | Overview page. Verify that your \$100 credits and 365 days of validity are active.

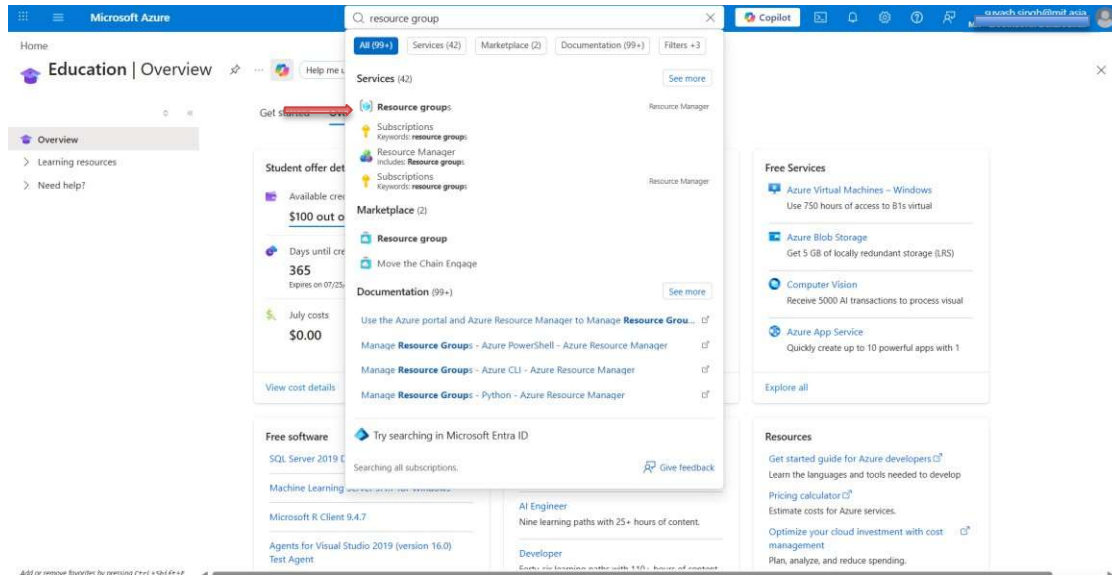


Step Image: Screenshot 2026-07-25 163915.png

2. Creating a Resource Group

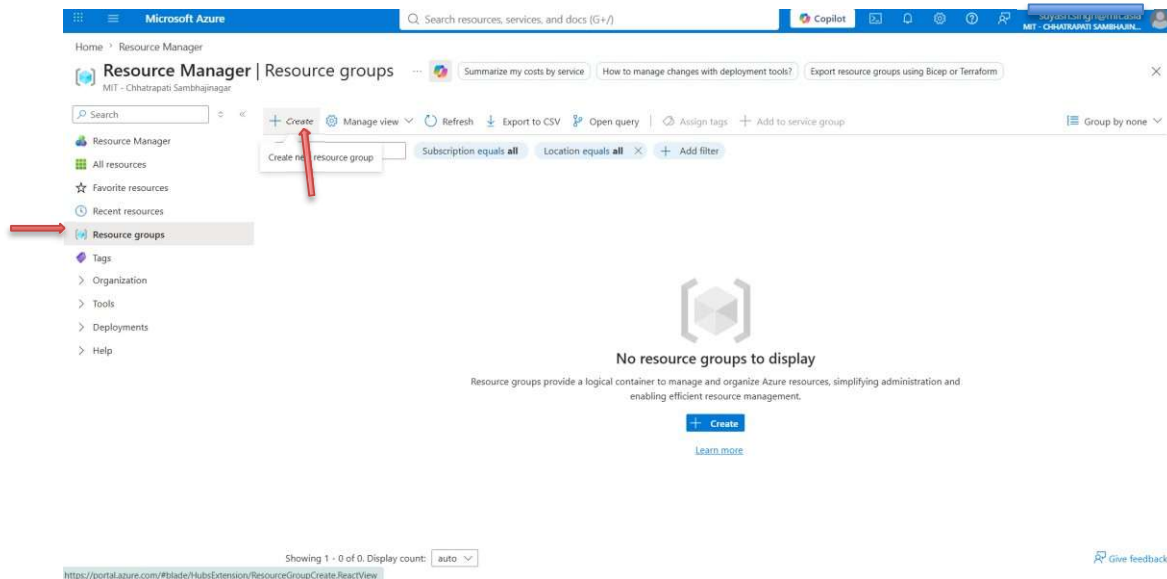
A Resource Group serves as a logical metadata folder that groups your Azure resources for billing, security, and administrative management.

- **Step 2.1:** Type 'resource group' in the portal's top global search box and click on the 'Resource groups' service.



Step Image: Screenshot 2026-07-25 164044.png

- **Step 2.2:** On the Resource Groups folder page, click the '+ Create' button on the toolbar.



Step Image: Screenshot 2026-07-25 164116.png

- **Step 2.3:** In the 'Basics' tab, set the Subscription to 'Azure for Students', name the Resource Group 'rg-group', and select a deployment Region (e.g., South India).

The screenshot shows the 'Create a resource group' page in the Microsoft Azure portal. The 'Basics' tab is selected. The form contains the following fields:

- Subscription ***: A dropdown menu with 'Azure for Students' selected.
- Resource group name ***: A text input field containing 'rg-group'.
- Region ***: A dropdown menu with '(Asia Pacific) South India' selected.

At the bottom of the form, there are three buttons: 'Previous', 'Next', and 'Review + create'.

Step Image: Screenshot 2026-07-25 164150.png

- **Step 2.4:** Click 'Next: Tags' to proceed (Tags are optional metadata; leave them empty).

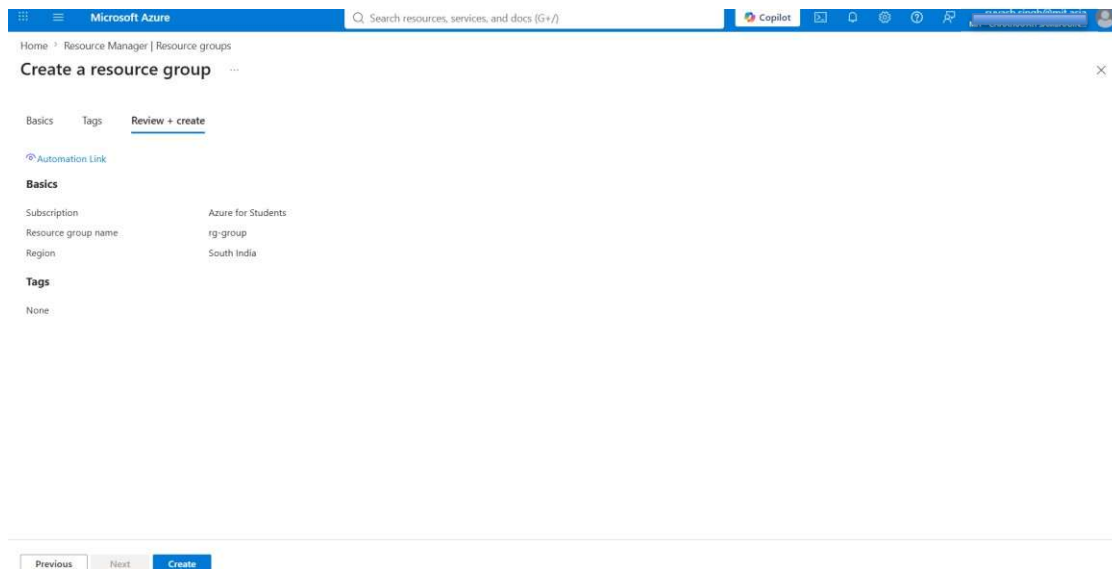
The screenshot shows the 'Create a resource group' page in the Microsoft Azure portal, now in the 'Tags' tab. The form contains the following fields:

Name	Value	Resource
		Resource group

At the bottom of the form, there are three buttons: 'Previous', 'Next', and 'Review + create'.

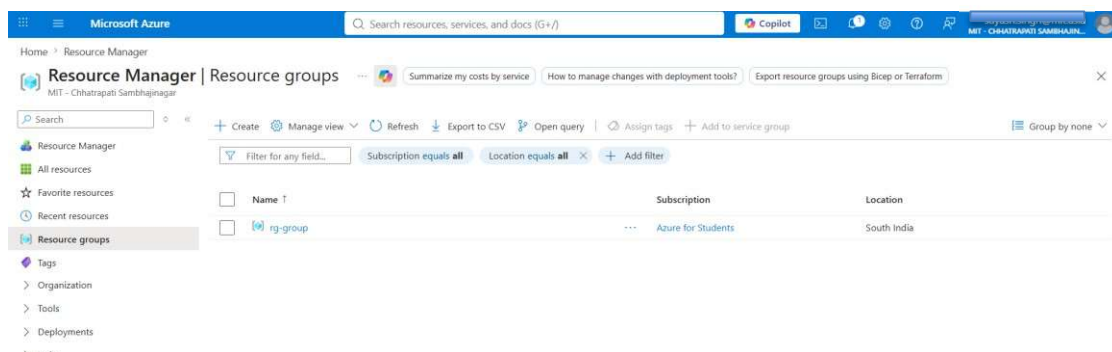
Step Image: Screenshot 2026-07-25 164205.png

- **Step 2.5:** Click 'Review + create'. Once validation succeeds, verify the parameters and click 'Create'.



Step Image: Screenshot 2026-07-25 164218.png

- **Step 2.6:** Confirm that your resource group 'rg-group' appears successfully in your list.

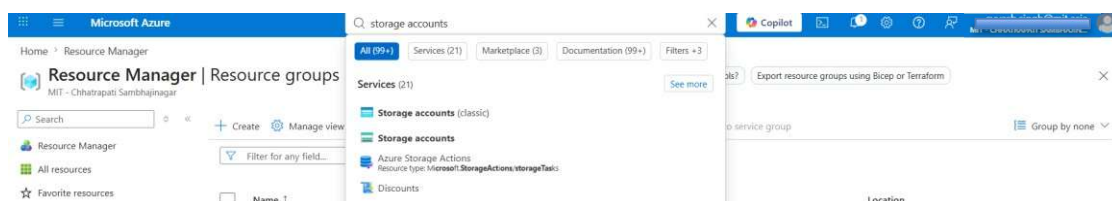


Step Image: Screenshot 2026-07-25 164247.png

3. Creating a Storage Account

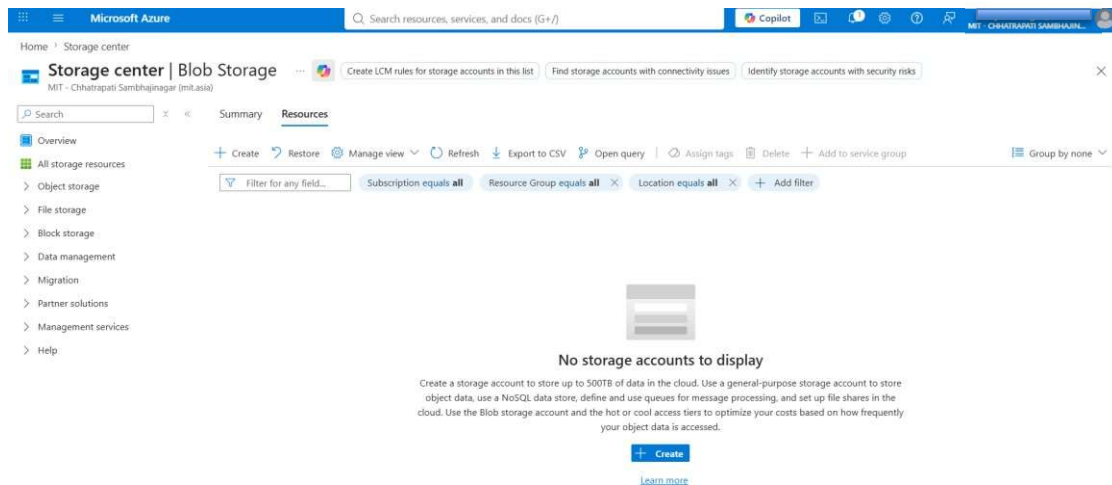
An Azure Storage Account contains all of your cloud storage resources: blob containers, files, queues, and tables.

- **Step 3.1:** Search for 'storage accounts' in the top search bar and click on the 'Storage accounts' service.



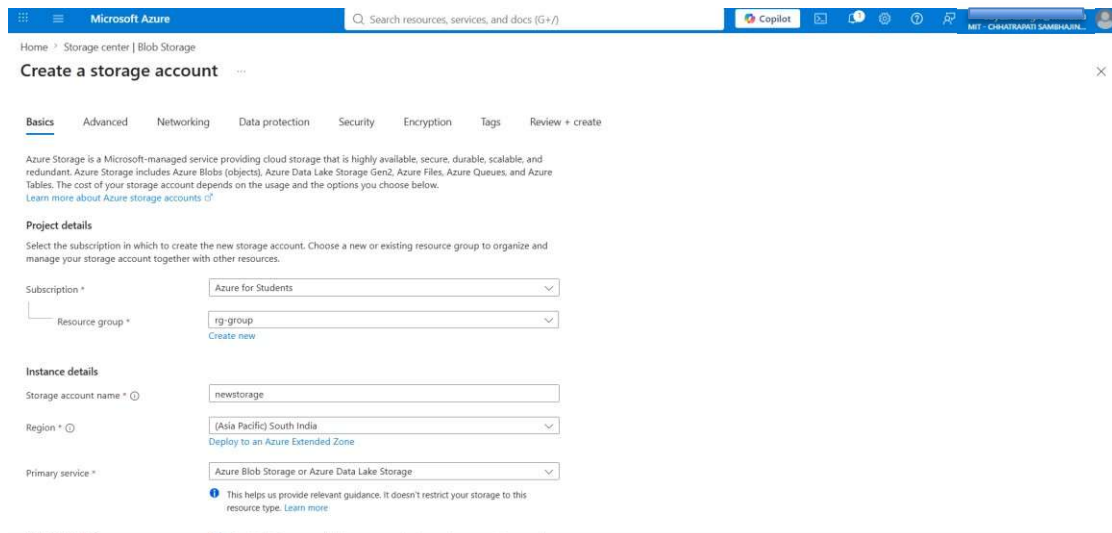
Step Image: Screenshot 2026-07-25 164414.png

- **Step 3.2:** Click the '+ Create' button to deploy a new storage instance.



Step Image: Screenshot 2026-07-25 164518.png

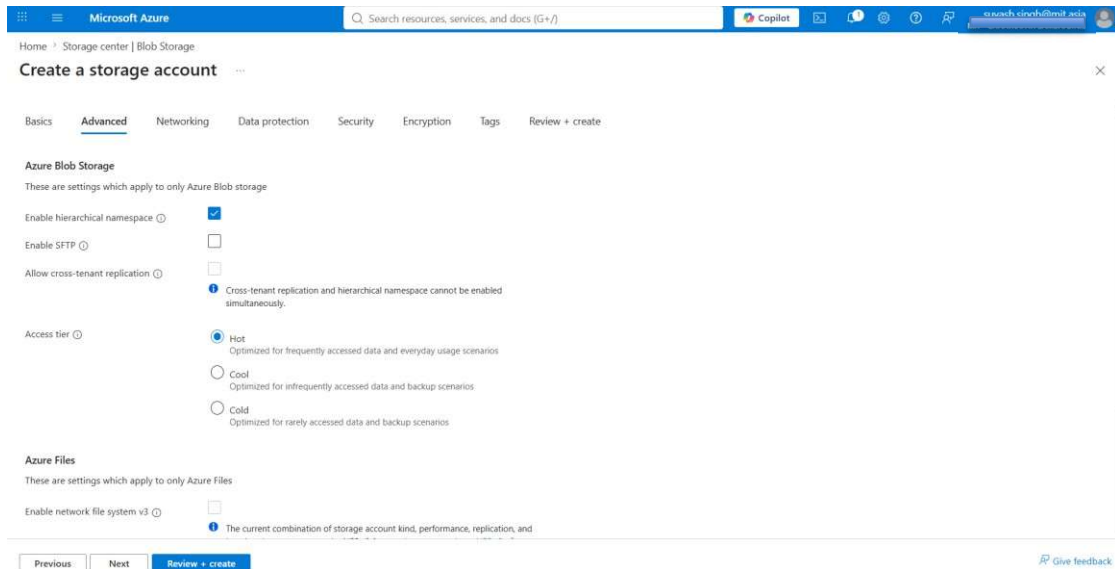
- **Step 3.3:** In the 'Basics' tab, select 'Azure for Students' subscription and the resource group 'rg-group'. Under 'Storage account name', type a unique name. Select a region, standard performance, and GRS or LRS redundancy.



Step Image: Screenshot 2026-07-25 164603.png

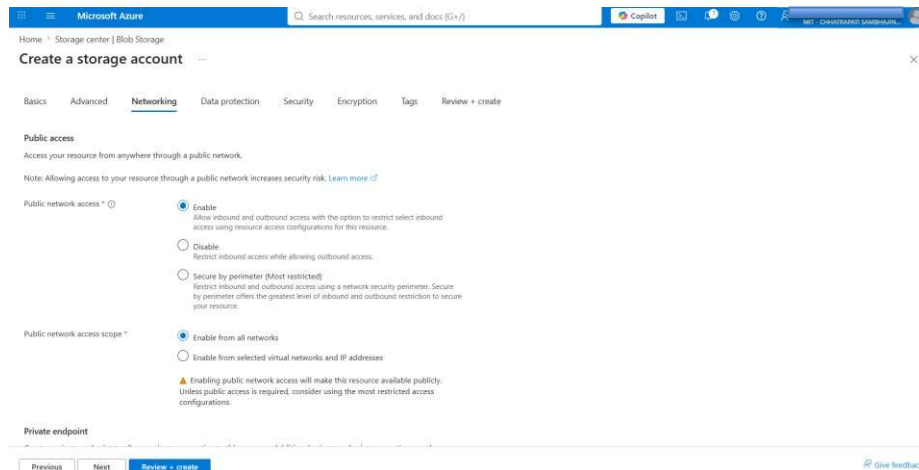
[Warning] Storage account names must be globally unique across all of Azure. If a common name like 'newstorage' is taken, try a unique name like 'mystorage879'. Also, you may need to adjust the deployment region based on quota availability.

- **Step 3.4:** Under the 'Advanced' tab, ensure 'Enable hierarchical namespace' is checked. This enables ADLS Gen2, which is ideal for big data analytics operations.



Step Image: Screenshot 2026-07-25 164621.png

- **Step 3.5:** Review other configuration tabs like Networking, Data protection, Security, Encryption, and Tags. Keep their default parameters.
- Networking Tab (Public access from all networks enabled):



Step Image: Screenshot 2026-07-25 164634.png

- Data Protection Tab (Soft delete enabled for safety):

Microsoft Azure

Home > Storage center > Blob Storage

Create a storage account

Basics Advanced Networking **Data protection** Security Encryption Tags Review + create

Azure Blob Storage
These are settings for recovery, tracking, and access control which apply to only Azure Blob Storage.

- ☐ **Enable point-in-time restore for containers**
Use point-in-time restore to restore one or more containers to an earlier state. If point-in-time restore is enabled, then versioning, change feed, and blob soft delete must also be enabled. [Learn more](#)
- ☒ **Enable soft delete for blobs**
Soft delete enables you to recover blobs and directories that were previously marked for deletion. [Learn more](#)
Days to retain deleted blobs
- ☒ **Enable soft delete for containers**
Soft delete enables you to recover containers that were previously marked for deletion. [Learn more](#)
Enabling soft delete for frequently overwritten data may result in increased storage costs. [Learn more](#)
Days to retain deleted containers
- ☐ **Enable versioning for blobs**
Use versioning to automatically maintain previous versions of your blobs. [Learn more](#)
Consider your workloads, their impact on the number of versions created, and the resulting costs. Optimize costs by automatically managing the data lifecycle. [Learn more](#)
- ☐ **Enable blob change feed**

[Previous](#) [Next](#) [Review + create](#)

[Give feedback](#)

Step Image: Screenshot 2026-07-25 164649.png

- Security and Encryption Tabs:

Microsoft Azure

Home > Storage center > Blob Storage

Create a storage account

Basics Advanced Networking Data protection **Security** Encryption Tags Review + create

Security
Configure security settings that impact your storage account.

- ☒ **Require secure transfer for REST API operations**
- ☐ **Allow enabling anonymous access on individual containers**
- ☒ **Enable storage account key access**
- ☐ **Default to Microsoft Data authorization in the Azure portal**
- Minimum TLS version**
- Permitted scope for copy operations (preview)**

Microsoft Defender for Storage
Microsoft Defender for Storage is an Azure native layer of security intelligence that detects potential threats of your storage accounts. It helps prevent the three major impacts on your data and workload: malicious file uploads, sensitive data exfiltration and data corruption.

- ☐ **Enable Defender for Storage**

[Previous](#) [Next](#) [Review + create](#)

[Give feedback](#)

Step Image: Screenshot 2026-07-25 164703.png

Microsoft Azure

Home > Storage center > Blob Storage

Create a storage account

Basics Advanced Networking Data protection Security **Encryption** Tags Review + create

Encryption type ☒ Microsoft-managed keys (BMAK) ☐ Customer-managed keys (CMK)

Enable support for customer-managed keys ☐ Blobs and files only ☐ All service types (blobs, files, tables, and queues)
This option cannot be changed after the storage account is created.

Enable infrastructure encryption ☐

[Previous](#) [Next](#) [Review + create](#)

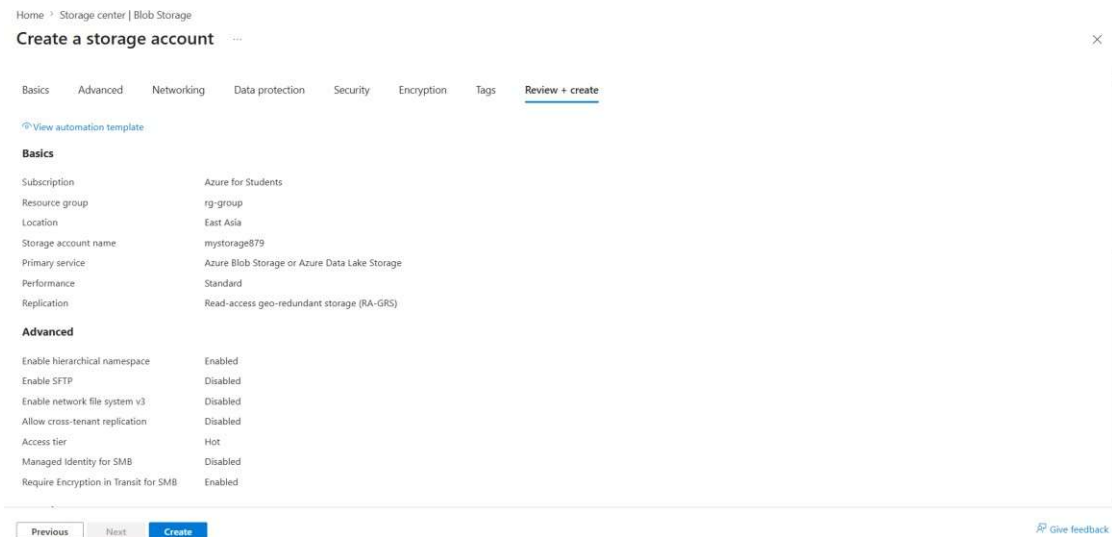
[Give feedback](#)

Step Image: Screenshot 2026-07-25 164718.png



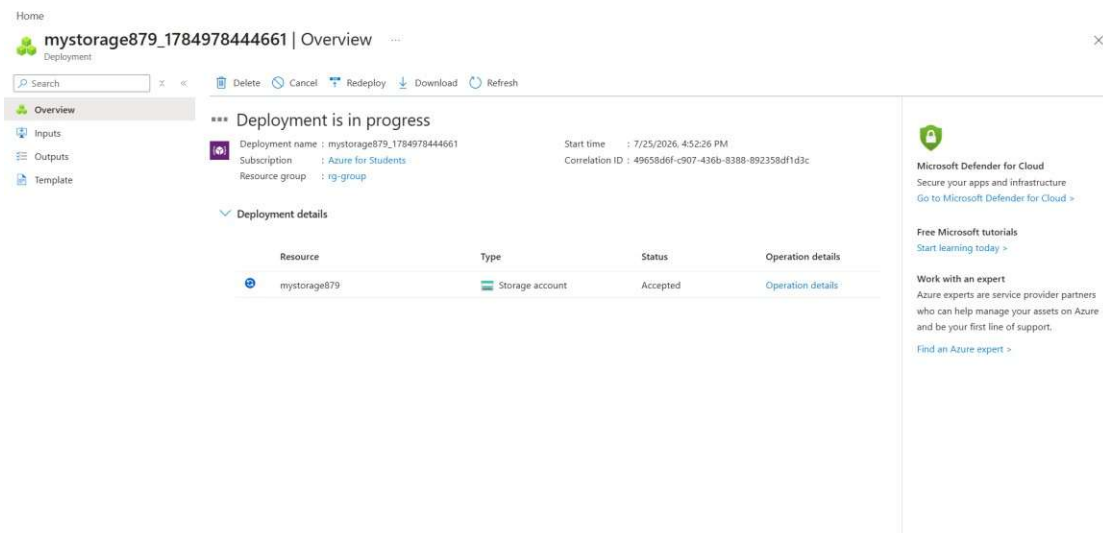
Step Image: Screenshot 2026-07-25 164751.png

- **Step 3.6:** In the 'Review + create' tab, click 'Create' after validation is successful. (Notice name changed to 'mystorage879' in region 'East Asia').



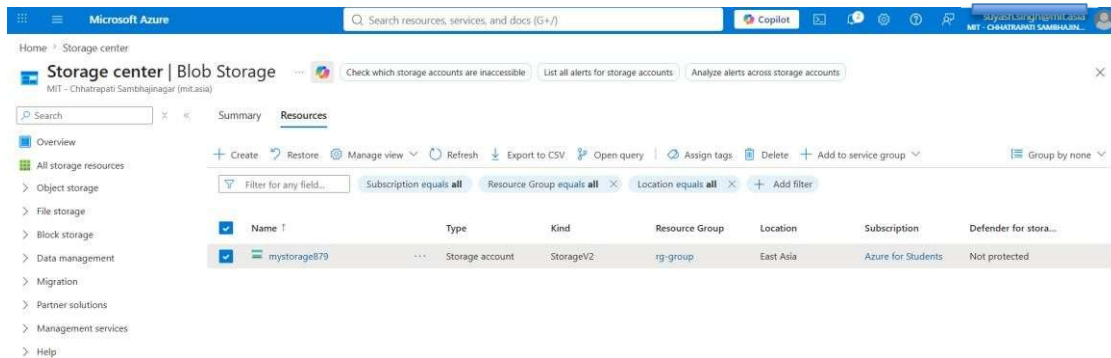
Step Image: Screenshot 2026-07-25 165158.png

- **Step 3.7:** Monitor the progress page which displays 'Deployment is in progress'.



Step Image: Screenshot 2026-07-25 165251.png

- **Step 3.8:** Once done, confirm that your storage account 'mystorage879' appears active in the storage accounts directory.

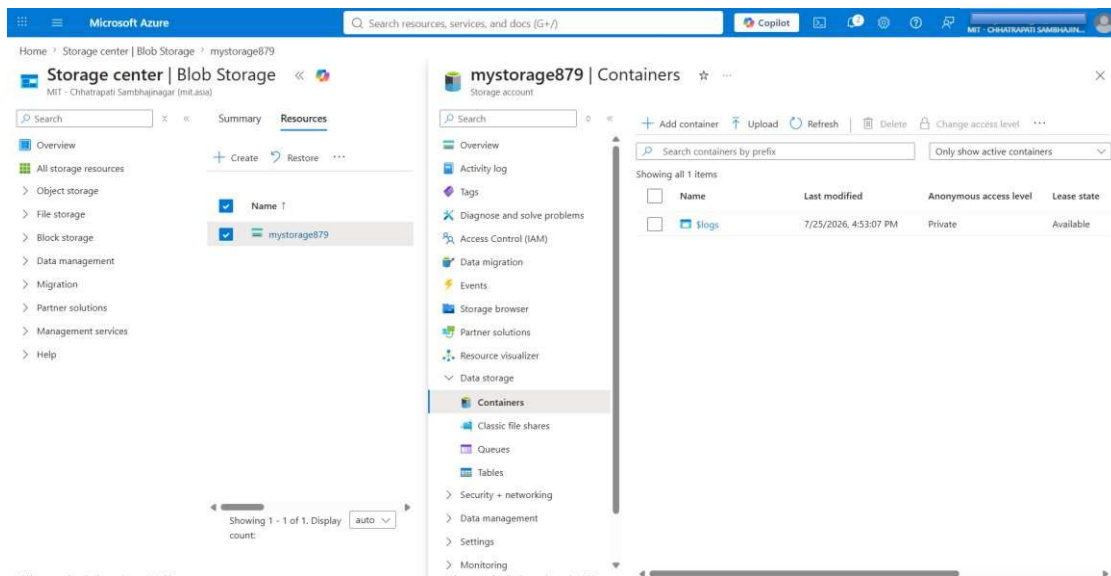


Step Image: Screenshot 2026-07-25 165436.png

4. Creating a Container

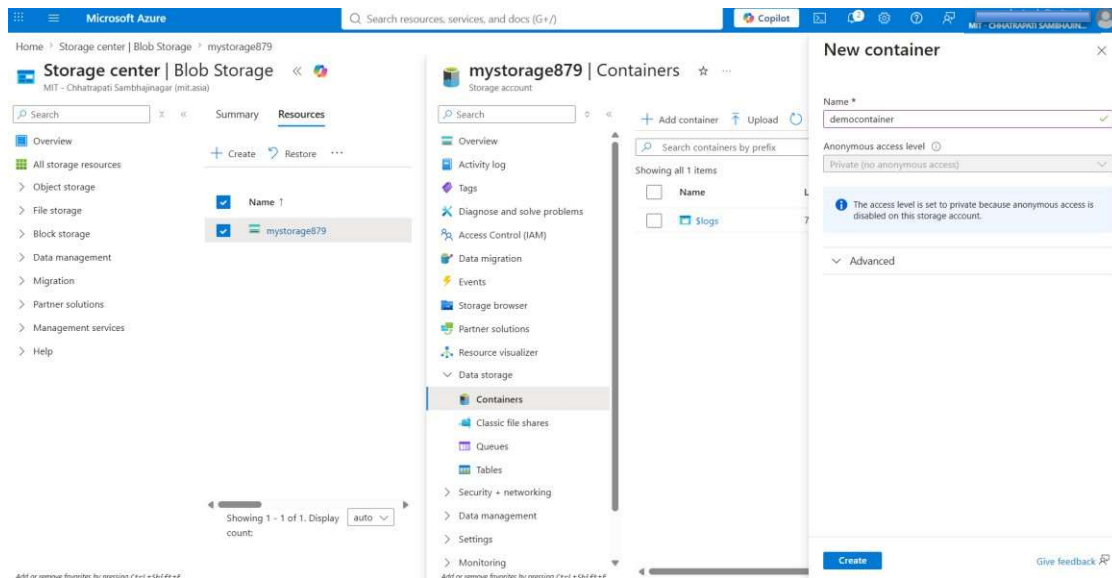
Azure Blob storage organizes data in containers, analogous to folders on a disk.

- **Step 4.1:** Open the 'mystorage879' storage account dashboard, find 'Containers' in the left menu under 'Data storage', and click '+ Add container'.



Step Image: Screenshot 2026-07-25 171346.png

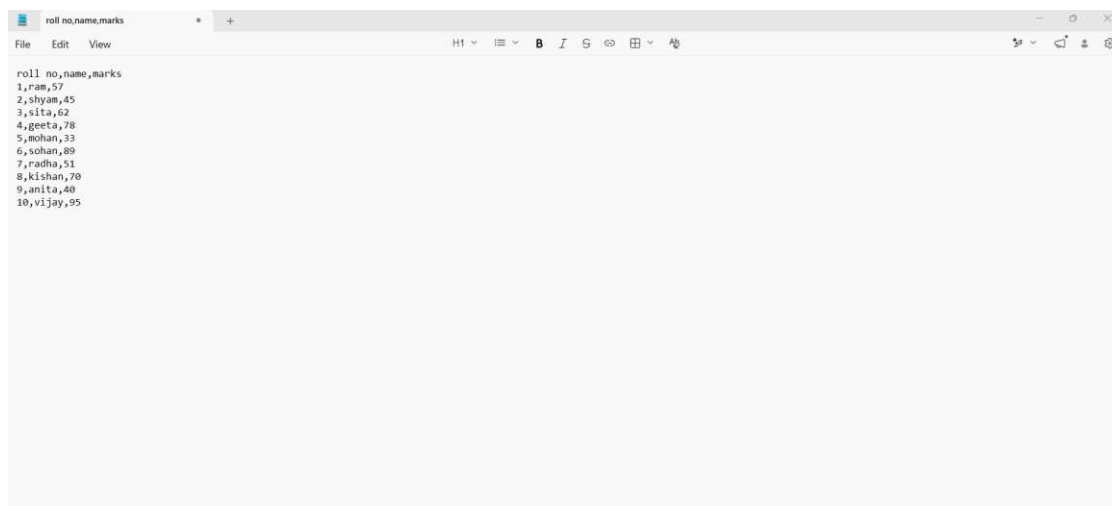
- **Step 4.2:** Name the container 'democontainer', leave the Access Level as 'Private', and click 'Create'.



Step Image: Screenshot 2026-07-25 171409.png

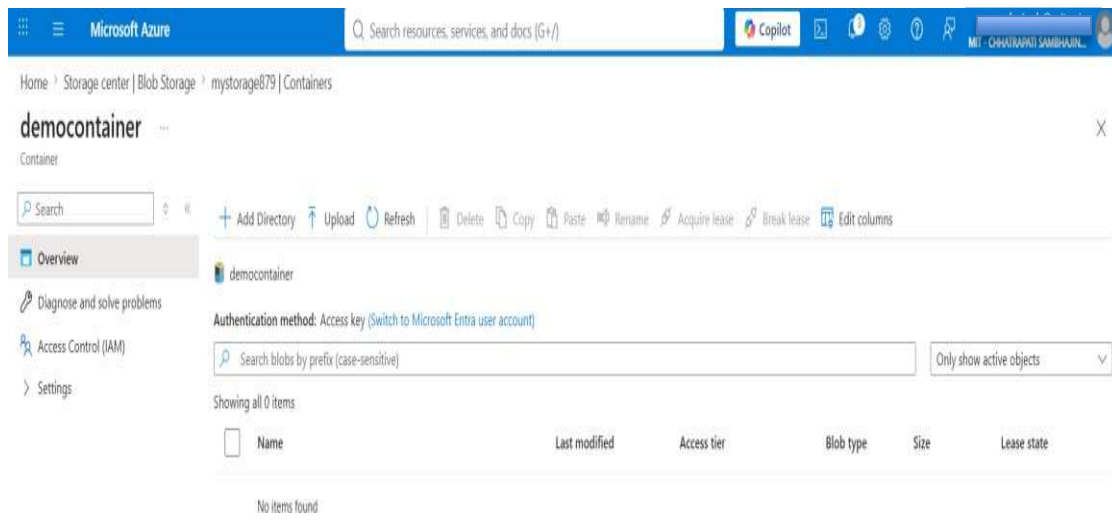
5. Preparing and Uploading a File

- **Step 5.1:** On your computer, open a text editor (Notepad) and write student data in CSV format. Save it locally as 'demo.csv'.



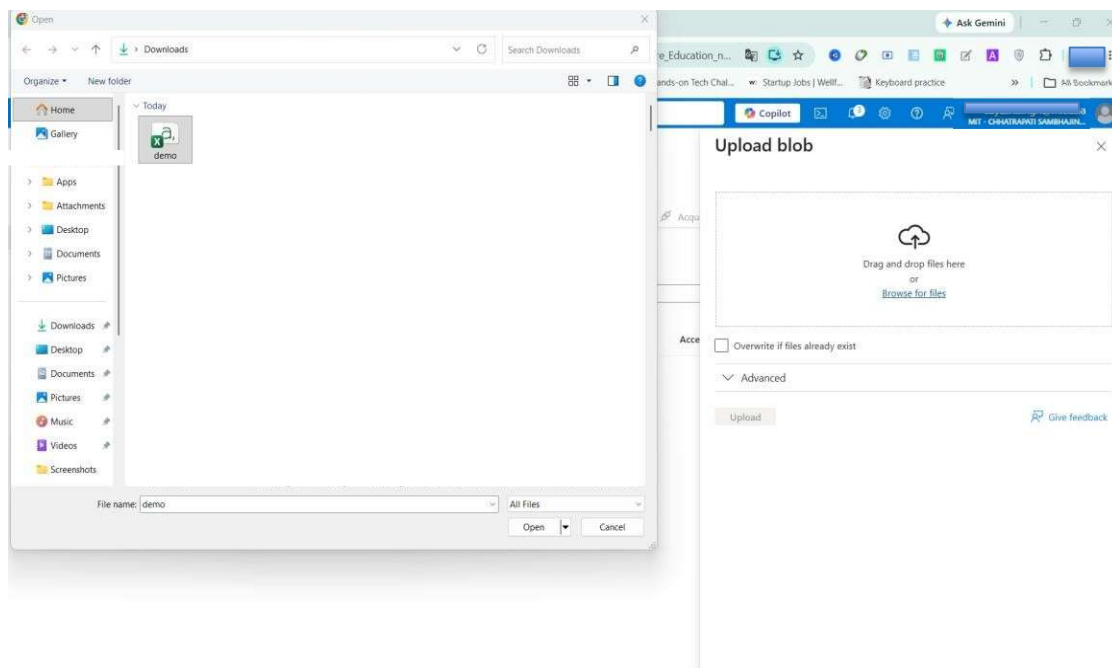
Step Image: Screenshot 2026-07-25 171117.png

- **Step 5.2:** Inside your container 'democontainer' in the portal, click the 'Upload' button on the toolbar.



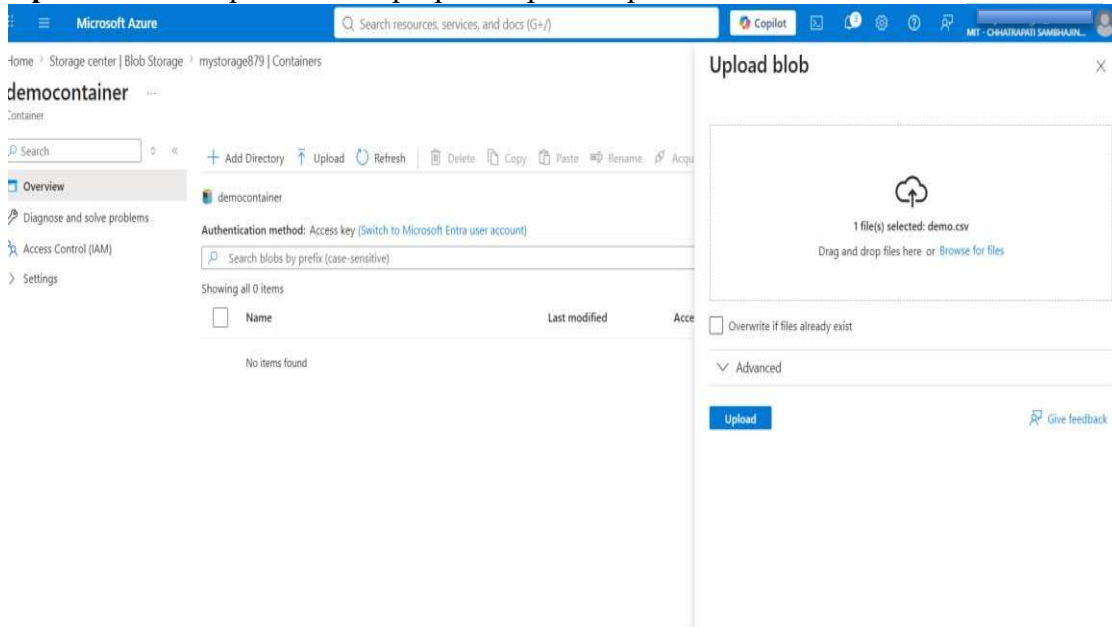
Step Image: Screenshot 2026-07-25 171427.png

- **Step 5.3:** In the panel, browse and select the 'demo.csv' file from your system downloads.



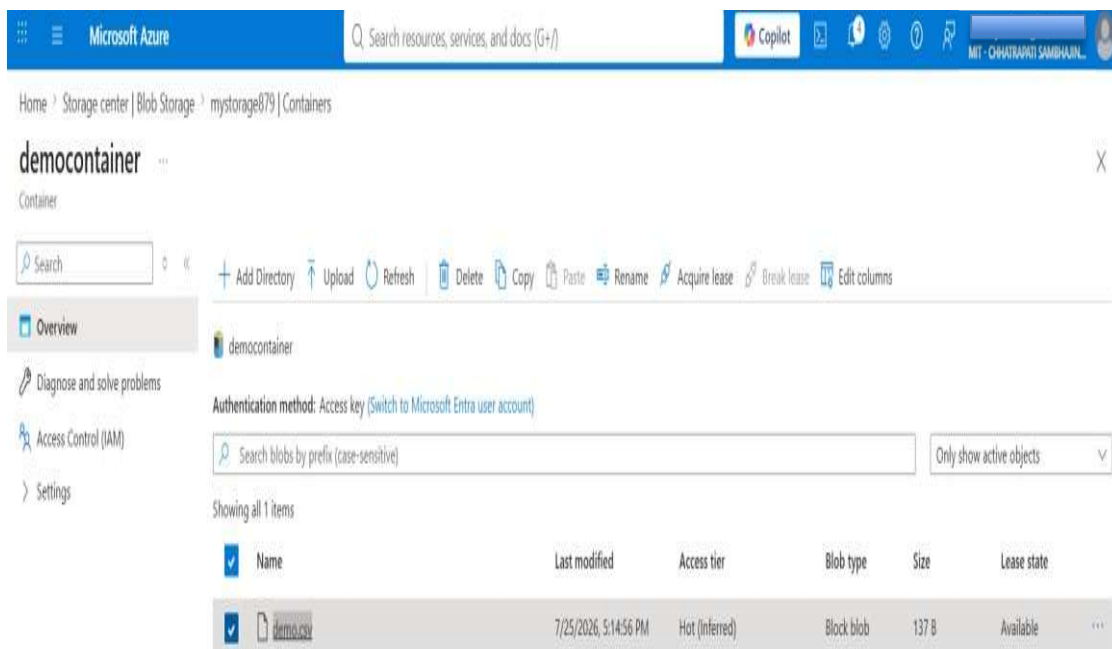
Step Image: Screenshot 2026-07-25 171442.png

- **Step 5.4:** Click 'Upload' in the properties pane to push the file to the cloud.



Step Image: Screenshot 2026-07-25 171453.png

- **Step 5.5:** Verify that 'demo.csv' is listed in the container's contents.



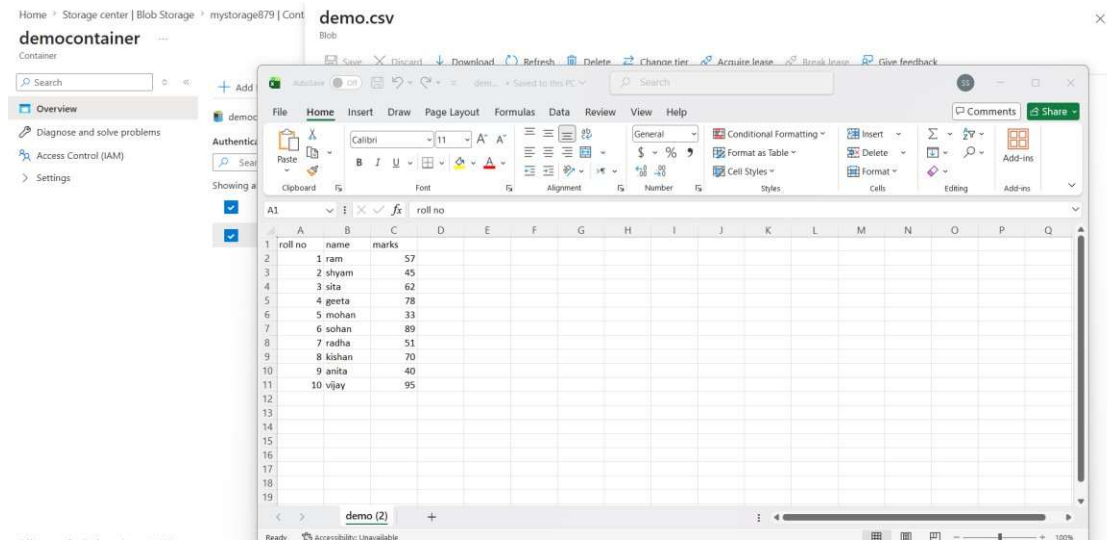
Step Image: Screenshot 2026-07-25 171552.png

6. Verifying the Uploaded File

You can verify the successfully uploaded data through multiple approaches:

Method 1: Local Application Verification

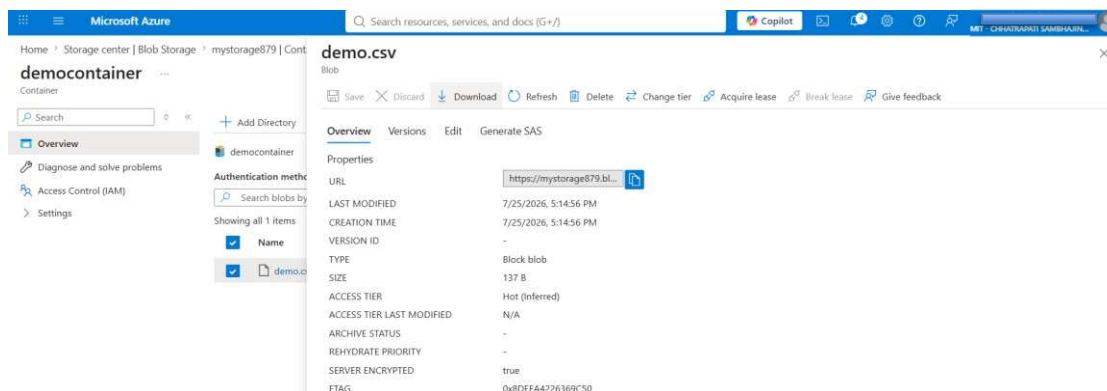
Open or double-click the original CSV file locally using Microsoft Excel to check column alignment (roll no, name, marks).



Step Image: Screenshot 2026-07-25 171636.png

Method 2: Metadata Inspection

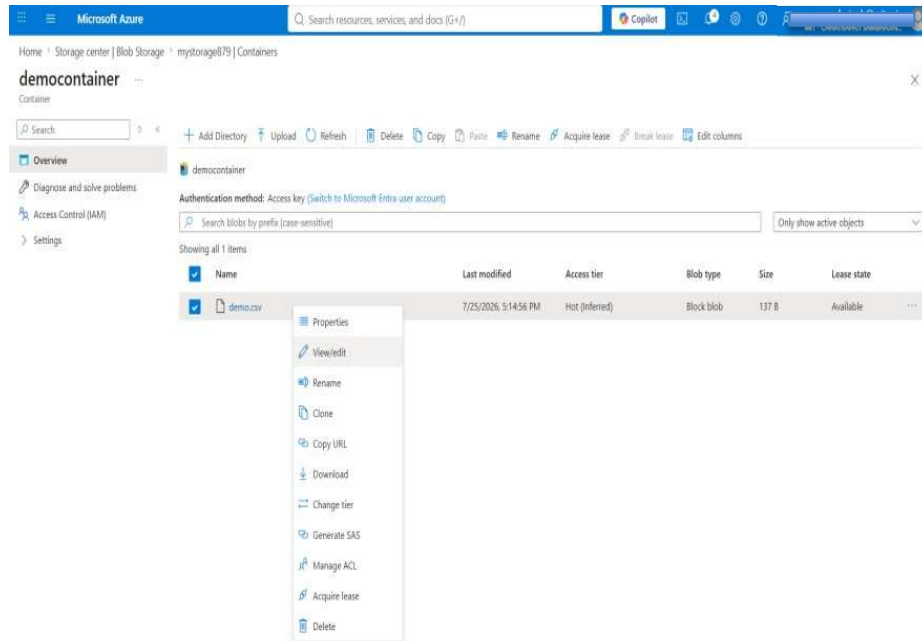
Click on 'demo.csv' in the container view. Under 'Overview', examine its properties: size (137 B), Blob type (Block blob), and copy the endpoint URL.



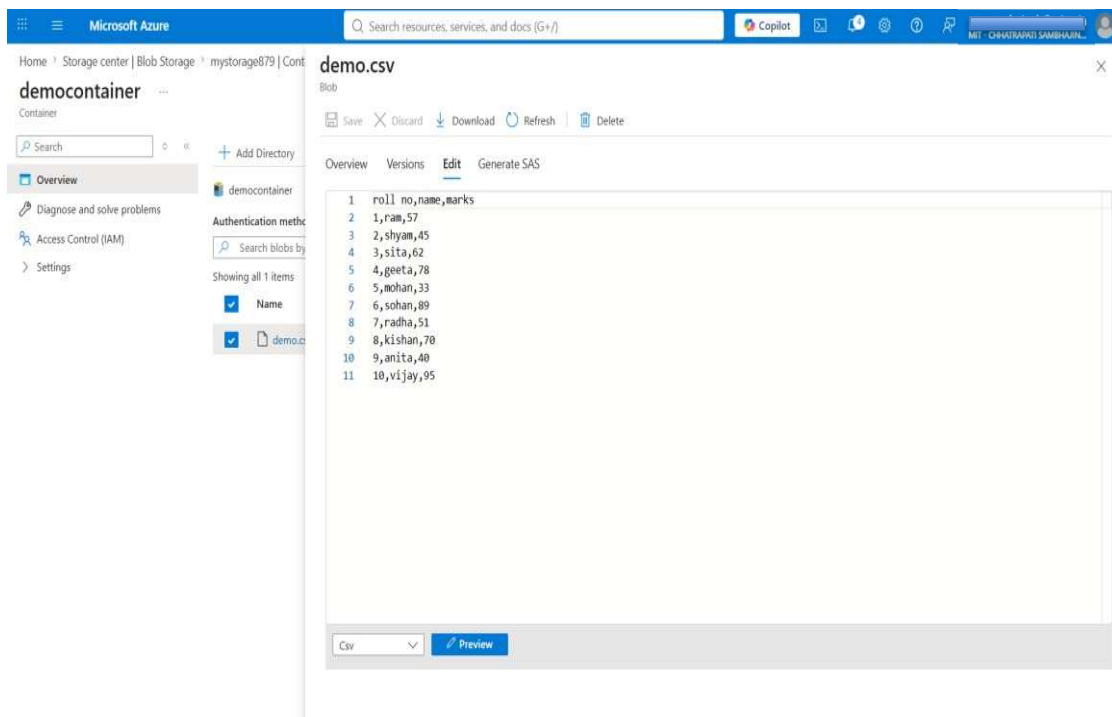
Step Image: Screenshot 2026-07-25 171612.png

Method 3: In-Portal Raw Text View

Click on the file options (...) and choose 'View/edit' to inspect the CSV text formatting directly in the cloud editor.



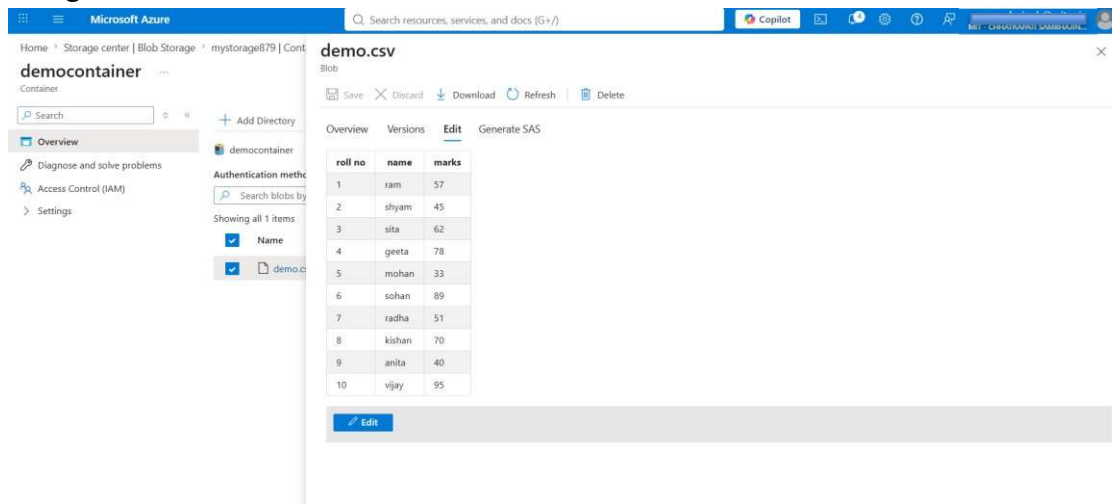
Step Image: Screenshot 2026-07-25 171658.png



Step Image: Screenshot 2026-07-25 171718.png

Method 4: Structured Cloud Preview

Click the 'Preview' button in the cloud editor. Azure parses the CSV and renders the student data in a clear grid/table.



Step Image: Screenshot 2026-07-25 171729.png

Practical Related Questions:

- What is object storage?
- Differentiate Amazon S3 and Azure Blob Storage.
- What is a storage bucket?

Conclusion: The practical was successfully completed by creating a cloud storage bucket/container and uploading a dataset in CSV/JSON format. Basic cloud storage operations such as uploading, downloading, and organizing files into folders were demonstrated using Azure Blob Storage. This practical provided hands-on understanding of how cloud storage enables scalable, reliable, and easily accessible data storage and management.